cXML Solutions Guide

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SAP Business Network for Trading Partners | 2602

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Catalog Upload Transaction

The cXML Catalog Upload transaction enables suppliers to upload and publish customer catalogs on SAP Business Network programmatically. The transaction provides an alternative to logging in to SAP Business Network to upload and publish customer catalogs interactively. Use it to distribute updated customer catalogs automatically whenever you change pricing or availability of products or services.

SAP Business Network supports any customer catalog in CIF, cXML, or BMEcat format. A CIF catalog can be either a text file saved with a `.cif` extension or an Excel file saved with a `.xls` extension. If you use the Catalog Upload transaction, the customer catalog cannot exceed 10 MB.

SAP Business Network allows only suppliers, not buying organizations, to upload customer catalogs.

Before uploading an Excel file as your customer catalog, be sure the file does not exceed 1 MB. Compress any Excel file that exceeds that limit before trying to upload it.

The transaction gives you an alternative to logging in to SAP Business Network to interactively upload and publish customer catalogs. You can use it to automatically distribute updated catalogs whenever you change pricing or availability of your products or services.

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Catalog Upload Transaction Overview

The Catalog Upload transaction consists of two cXML documents, `CatalogUploadRequest` and `Response`.

The purpose of these documents is as follows:

cXML Document	Purpose
`CatalogUploadRequest`	Sent by suppliers to upload a customer catalog. It contains the customer catalog as an attachment and specifies whether the customer catalog is new or an update, and whether to automatically publish it after upload.
`Response`	Sent by SAP Business Network to acknowledge the receipt of a `CatalogUploadRequest`.

These documents use the standard cXML transport mechanism to travel between your cXML-enabled catalog-management application and SAP Business Network:

1. Your catalog-management application opens an HTTPS connection to SAP Business Network and perform a POST to send the `CatalogUploadRequest` document.
2. SAP Business Network receives the document, validates it, and parses it.

3. After verifying your credential, Ariba sends a `Response` document through the same HTTPS connection.

4. Your application receives the `Response` document and closes the HTTPS connection.

You can receive further status regarding the customer catalog through email or cXML.

`CatalogUpoadRequest` Prerequisites

The `CatalogUploadRequest` transaction has requirements for your SAP Business Network account, cXML-enabled applications, and cXML document authentication. It also requires valid customer catalogs.

SAP Business Network Account

You can have a standard or enterprise supplier account on SAP Business Network to load the `CatalogUploadRequest` transaction. However, if you have a standard supplier account, you must have an active buyer relationship.

CXML-Enabled Application

You must have a cXML-enabled application that can initiate HTTPS POST operations.

cXML-enabled applications must support sending and receiving the `ProfileRequest` transaction. This transaction is important for the Catalog Upload transaction because:

- You will use it to find out URL to which to post the `CatalogUploadRequest` document.
- SAP Business Network will use it to find out the URL to which to post `StatusUpdateRequest` documents to you for later customer catalog-status notification.

CXML-Document Authentication

You must have configured your SAP Business Network account and your cXML-enabled application to authenticate received cXML documents.

You must be the account administrator to change your account settings, such as cXML authentication method.

Valid Catalogs

You must have valid customer catalogs. Customer catalogs can be in CIF 2.1, CIF 3.0, cXML, BMEcat, or Excel format.

To debug your customer catalogs as you develop them, use your SAP Business Network account to upload them. SAP Business Network automatically validates them and flags any syntactic errors. Your customer catalogs are validated when you use the `CatalogUploadRequest` transaction, but solving catalog problems is easier in interactive sessions. Interactive sessions also allow you to use the order tester to exercise customer catalogs and generate test purchase orders. Compress (zip) large customer catalogs before uploading.

ℹ Note

You can't use the `CatalogUploadRequest` transaction to publish new customer catalogs to buyers. You can publish a customer catalog using the `CatalogUploadRequest` transaction only if there is a previous version of the customer catalog already existing on SAP Business Network.

However, in the case of BMEcat catalogs, you can use the `CatalogUploadRequest` transaction to publish both new catalogs as well as updates to existing catalogs to CMS-enabled buyers. These catalogs are not validated by SAP Business Network and are automatically published to the CMS in the buyer's SAP Ariba solution. You do not have to log on to your account and manually publish these catalogs.

For Information About...	See...
Creating CIF and cXML catalogs	Customer Catalog Format Reference
Creating Excel format catalogs	Creating and Managing Customer Catalogs
Uploading, validating, and testing catalogs	Creating and Managing Customer Catalogs

`CatalogUploadRequest` Document

A `CatalogUploadRequest` document contains `To` and `CatalogUploadRequest`, and requires you to attach your valid customer catalog.

`To` Element

The `To` element can specify either SAP Business Network or the buying organization.

If you address the document to SAP Business Network, SAP Business Network performs generic customer catalog validation, not buyer-specific customer catalog validation. If you address the document to a buying organization, SAP Business Network performs validation against the customer's catalog rules.

`CatalogUploadRequest` Element

`CatalogUploadRequest` has an `Operation` attribute that specifies the type of upload to perform. It also contains several elements that detail aspects of the request.

The `Operation` attribute is one of:

- `new`: Uploads a new customer catalog. A customer catalog with the same name must not exist.
- `update`: Overwrites an exiting customer catalog. A customer catalog with the same name must exist.

`CatalogName` Element

`CatalogName` specifies the name of the uploaded customer catalog. This value is the user-visible name, not the file name of the customer catalog.

`CatalogName` has an `xml:lang` attribute that specifies the language used for the customer catalog name. Language codes are defined in the XML 1.0 Specification (at http://www.w3.org/TR/1998/REC-xml-19980210.html). In the most common case, this includes an ISO 639 Language Code and, optionally, an ISO 3166 Country Code separated by a hyphen. The recommended cXML language code format is xx[-YY[-zzz]*] where xx is an ISO 639 Language code, YY is an ISO 3166 Country Code, and zzz is an IANA or private subcode for the language in question. Again, use of the Country Code is always recommended. By convention, the language code is lowercase and the country code is uppercase. This is not required for correct matching of the codes.

Description

`Description` briefly describes the customer catalog contents. Buying organizations can search and view this information.

`Description` has an `xml:lang` attribute that specifies the language of the description text.

`Attachment` Element

`Attachment` specifies the URL of the attached customer catalog. It contains one URL element with the scheme “cid:”.

Related Information

Receiving the Response

`Commodities` Element

`Commodities` specifies the top-level commodity codes for the items in your customer catalog. Buying organizations use these codes to search for new customer catalogs. It contains one or more `CommodityCode` elements.

Use **two-digit** UNSPSC (United Nations Standard Products and Services Code) segment codes.

Related Information

Recommended Coding Systems

`AutoPublish` Element

`AutoPublish` automatically publishes the customer catalog to buying organizations after upload.

You can automatically publish only if the following requirements are met:

- A previous version of the customer catalog exists in your account and you are performing an `update` operation.
- The previous version is in the “published” state. It must have been published private (with a list of buying organizations) or public.
- The `CatalogUploadRequest` (the `To` element) is addressed to one of your customers, not SAP Business Network.

`AutoPublish` has an Enabled attribute that specifies whether to automatically publish the customer catalog:

- `true`—Publishes the customer catalog. It must be an update to a previously published customer catalog.
- `false`—Does not publish the customer catalog. You can log on to your account and manually publish the customer catalog.

ℹ Note

BMEcat catalogs are automatically published after upload irrespective of whether the `AutoPublish` element in the `CatalogUploadRequest` cXML document is set to `true` or `false`.

`Notification` Element

`Notification` sends customer catalog-status notifications through email or cXML POST.

For examples of these messages, see Receiving Later Customer Catalog Status.

`Notification` contains one or both of the `Email` element and `URLPost` elements.

`Email` Element

`Email` specifies the mailbox to which SAP Business Network emails status messages. You can use only one `Email` element, and it can contain only one email address.

`URLPost` Element

`URLPost` specifies whether SAP Business Network sends customer catalog status messages as cXML `StatusUpdateRequest` documents.

The URL destination of the `StatusUpdateRequest` is determined by your Website's response to SAP Business Network's `ProfileRequest` transaction.

`URLPost` has an Enabled attribute that specifies whether SAP Business Network sends customer catalog-status notifications through `StatusUpdateRequest`:

- `true`: Enables this feature.
- `false`: Disables this feature.

Related Information

Profile Transaction

Attaching Your Customer Catalog

Send your customer catalog attached to the `CatalogUploadRequest` document as MIME attachment. Large customer catalogs must be zipped to compress them before uploading.

Use a MIME Envelope

Include the customer catalog file in the `CatalogUpdateRequest` as a Multipurpose Internet Mail Extensions (MIME) attachment. cXML contains only references to external MIME parts sent within one multipart MIME envelope.

The referenced customer catalog file must reside within a multipart MIME envelope with the cXML document. A cXML requirement for this envelope (over the basics described in RFC 2046 "Multipurpose Internet Mail Extensions Part Two: Media Types") is the inclusion of Content-ID headers with the attached file.

ℹ Note

The cXML specification allows attachments to reside outside of the MIME envelope, but the Catalog Upload transaction does not support that attachment method.

The `Attachment` element contains only a reference to the external MIME part of the attachment. `Attachment` contains a single URL with the scheme "cid:". Do not use any encoding for the customer catalog, except the appropriate character set encoding (for example, UTF-8). Do not base64 encode the customer catalog.

For more information about attachments in cXML, see the discussion of the `Attachment` element in the *cXML reference guide* at http://www.cxml.org .

Compressing Customer Catalogs

Before attaching customer catalogs larger than 3 MB, compress them with a Zip utility, such as WinZip.

Customer catalog files must have a `.xml`, `.cif`, `.xls`, or `.zip` file extension.

Example of Sending a `CatalogUploadRequest` Document

The example of sending a `CatalogUploadRequest` document contains two documents enclosed in a MIME envelope: a `CatalogUploadRequest` document and a CIF 3.0 catalog.

```
--kdflkajfdksadjfklasdjfkljdfdsfdkf
Content-type: text/xml; charset=UTF-8
Content-ID: <part0.PC028.975529413484@saturn.workchairs.com>
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE cXML SYSTEM "http://xml.cxml.org/schemas/cXML/1.2.023/cXML.dtd">
<cXML timestamp="2011-12-28T16:56:03-08:00" payloadID="123456669138--1234567899555556789@10.10.83.3
        <Header>
                <From>
                        <Credential domain="DUNS">
                                <Identity>123456789</Identity>
                        </Credential>
                </From>
                <To>
                        <Credential domain="NetworkID">
                                <Identity>AN01000000001</Identity>
                        </Credential>
                </To>
                <Sender>
                        <Credential domain="DUNS">
                                <Identity>123456789</Identity>
                                <SharedSecret>abracadabra</SharedSecret>
                        </Credential>
                        <UserAgent>MyHomemadeCatalogManager</UserAgent>
                </Sender>
        </Header>
        <Request>
                <CatalogUploadRequest operation="update">
                        <CatalogName xml:lang="en">Winter Prices</CatalogName>
                        <Description xml:lang="en">This catalog contains our
                                premiere-level prices for office chairs and other durable
                                furniture.</Description>
                        <Attachment>
                                <URL>cid: part2.PC028.975529413154@saturn.workchairs.com</URL>
                        </Attachment>
                        <Commodities>
                                CommodityCode>52</CommodityCode>
                        </Commodities>
                        <AutoPublish enabled="true"/>
                        <Notification>
                                <Email>judy@workchairs.com</Email>
                                <URLPost enabled="true"/>
```

```
                        </Notification>
                </CatalogUploadRequest>
        </Request>
</cXML>
--kdflkajfdksadjfklasdjfkljdfdsfdkf
Content-type: text/plain; charset=US-ASCII
Content-Disposition: attachment; filename=PremiereCatalog.cif
Content-ID: <part2.PC028.975529413154@saturn.workchairs.com>
Content-length: 364
CIF_I_V3.0
LOADMODE: F
CODEFORMAT: UNSPSC
CURRENCY: USD
SUPPLIERID_DOMAIN: DUNS
ITEMCOUNT: 3
TIMESTAMP: 2006-01-15 15:25:04
DATA
942888710,34A11,C11,"Eames Chair",11116767,400.00,EA,3,"Fast MFG",,,400.00
942888710,56A12,C12,"Eames Ottoman",11116767,100.00,EA,3,"Fast MFG",,,100.00
942888710,78A13,C13,"Folding Chair",11116767,25.95,EA,3,"Fast MFG",,,25.95
ENDOFDATA
--kdflkajfdksadjfklasdjfkljdfdsfdkf--
```

Receiving the Response

After you send a `CatalogUploadRequest`, SAP Business Network replies with a standard cXML `Response` document.

Example

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE cXML SYSTEM "http://xml.cxml.org/schemas/cXML/1.2.024/cXML.dtd">
<cXML payloadID="980306507433-6714998277961341012@10.10.83.39"
                        timestamp="2011-01-23T19:21:47-08:00">
        <Response>
                <Status code="201" text="Accepted">The catalog upload request is processing</Statu
        </Response>
</cXML>
```

The following table lists possible status codes:

Status Code	Meaning
200 Success	The customer catalog-upload request succeeded.
201 Accepted	The customer catalog-upload request is processing.
461 Bad Commodity Code	The commodity code you assigned to the customer catalog is invalid.
462 Notification Error	No notification method (email or URL) provided.
463 Bad Catalog Format	The zip file is invalid.
464 Bad Catalog	No customer catalog is attached, or more than one is attached.
465 Duplicate Catalog Name	The name of the customer catalog exists.
466 No Catalog to Update	The customer catalog to be updated does not exist.
467 Publish Not Allowed	You attempted to publish a customer catalog that was not previously published.
468 Catalog Too Large	The size of the customer catalog exceeds the 10 MB limit.

Status Code	Meaning
469 Bad Catalog Extension	The file name of the customer catalog must have .cif, .xml, or .zip extensions.
470 Catalog Has Errors	The message is the status of the customer catalog. (HasErrors)
499 Document Size Error	The cXML document is too large.
561 Too Many Catalogs	You cannot upload more than a specific number of customer catalogs per hour.
562 Publish Disabled	Customer catalog publishing is temporarily unavailable due to scheduled maintenance. It will be back online by the specified date and time.
563 Catalog Validating	You attempted to update a customer catalog before validation finished on a previous version of the customer catalog.
564 Upload Disabled	Customer catalog uploading is temporarily unavailable due to scheduled maintenance. Try again later.

For other status codes, see the *cXML reference guide* at http://www.cxml.org .

Receiving Later Customer Catalog Status

If you include the `Notification` element to request later customer catalog-status notification, SAP Business Network sends a message when the customer catalog reaches its final status.

The possible final customer catalog states are:

State	Meaning
`Validated`	The customer catalog contains no syntax errors.
`BadZipFormat`	The zip format is incorrect.
`HasErrors`	The customer catalog contains syntax errors, and it cannot be published.
`Published`	The customer catalog has been published (private or public).

This notification does not appear in your SAP Business Network Inbox. You receive it only through email or URLPost.

Email Notification of Customer Catalog Status

If you specify an email mailbox to receive customer catalog status updates in `CatalogUpdateRequest`, then SAP Business Network sends status notifications there.

The following example shows an email status notification:

```
From: SAPBusinessNetworkAdmin@sapariba.com
To: judy@workchairs.com
Subject: Status of CatalogUploadRequest for Winter Prices from the SAP Business Network
Catalog Name: Winter Prices
Status: Published
Payload ID: 123456669138--1234567899555556789@10.10.83.39
Please sign in to your SAP Business Network account to view or edit the catalog file.
Thank you,
SAP Business Network Customer Service
```

URLPost Notification of Customer Catalog Status

If you specify URLPost to receive customer catalog status updates in CatalogUpdateRequest, then SAP Business Network sends a cXML message to the URL you specified.

The following example shows a StatusUpdateRequest notification sent by SAP Business Network:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE cXML SYSTEM "http://xml.cxml.org/schemas/cXML/1.2.014/cXML.dtd">
<cXML timestamp="2001-01-23T18:39:44-08:00" payloadID="980303984882--3544419350291593786@10.10.83.3
        <Header>
                <From>
                        <Credential domain="NetworkID">
                                <Identity>AN01000000001</Identity>
                        </Credential>
                </From>
                <To>
                        <Credential domain="DUNS">
                                <Identity>123456789</Identity>
                        </Credential>
                </To>
                <Sender>
                        <Credential domain="NetworkID">
                                <Identity>AN01000000001</Identity>
                                <SharedSecret>abracadabra</SharedSecret>
                        </Credential>
                        <UserAgent>ANValidator</UserAgent>
                </Sender>
        </Header>
        <Request>
                <StatusUpdateRequest>
                        <DocumentReference
                                payloadID="123456669131-1234567899555556789@10.10.83.39">
                        </DocumentReference>
                        <Status text="Success" code="200">
                                Validated
                        </Status>
                </StatusUpdateRequest>
        </Request>
</cXML>
```

The possible status codes are:

Status Code	Meaning
200 Success	The customer catalog-upload request succeeded.
463 Bad Catalog Format	The zip file is invalid.
470 Catalog Has Errors	The message is the status of the customer catalog. (HasErrors)